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De Novo Construction of Substituted Terephthalates via Phosphine Catalyzed Domino Benzannulation Reactions

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Abstract: The robustness of phosphine catalysis enabling the domino benzannulation reactions of allenoates with enamines is described. A broad array of substituted terephthalates were delivered under simple and practical reaction conditions. Furthermore, the reaction could be carried out on a gram scale with higher yield, and various conversions of the terephthalate products demonstrate the versatility of this transformation.

Keywords: Terephthalates; Benzannulation; Allenoates; Enamines; Phosphine

Terephthalate structure commonly occurs in molecules within a diverse array of biological and pharmaceutical activities, as well as finding significant applications in polymer materials.[1] It is noteworthy that the global consumption of polyethylene terephthalate (PET) is estimated to be 108 million tons in 2020, which will utilize immense amounts of fossil oil.[2] In addition, introduction of terephthalate-containing systems allows increasing structural novelty and adding its value as a versatile building unit in pharmacologically relevant compounds (Figure 1).[3] Given its great prominence, seeking for efficient and selective methodologies to generate substituted terephthalates has remained a perennial interest. In principle, a conjugated diene and an alkene may enable direct access to *p*-Xylene via Diels-Alder reaction and dehydrogenation sequence, which need further oxidation to afford terephthalate.[4] Pleasantly, Xu's group proposed a synthetic route to diethyl terephthalate (DET), using biomass-derived muconic acid, ethanol, and ethylene as the reactants, through a cascade process combining esterification, Diels-Alder cycloaddition and dehydrogenation (Scheme 1a).[5] Whereas, more rigorous conditions are still required to proceed this protocol. Alternatively, transition-metal-catalyzed methods have long proven to be valuable for transferring functional group to precursor substrates.[6] For instance, Ritter's group showcased Ru-catalyzed C–H amination, leading to amino-substituted terephthalate (Scheme 1a).[7] Benzannulation was also pursued vigorously as a straightforward approach to polysubstituted aromatic compounds.[8] Among of those, transition-metal catalysts have demonstrated great potentials to carry out the preparation of functionalized benzenes. In recent advances, Moret's group achieved [2+2+2] cyclotrimerization with the use of Ni(0) complexes, selectively affording carboxylate substituted terephthalates from terminal alkynes (Scheme 1a).[9] Although a small number of such methods have been exploited, multistep strategies and restriction on the substitution

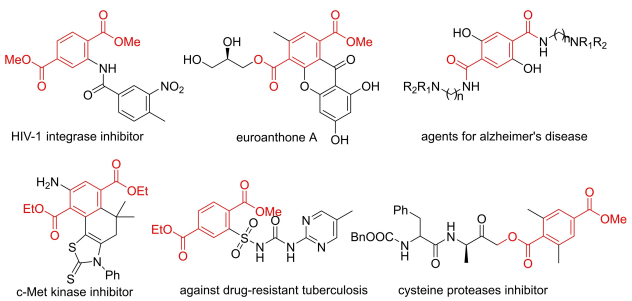
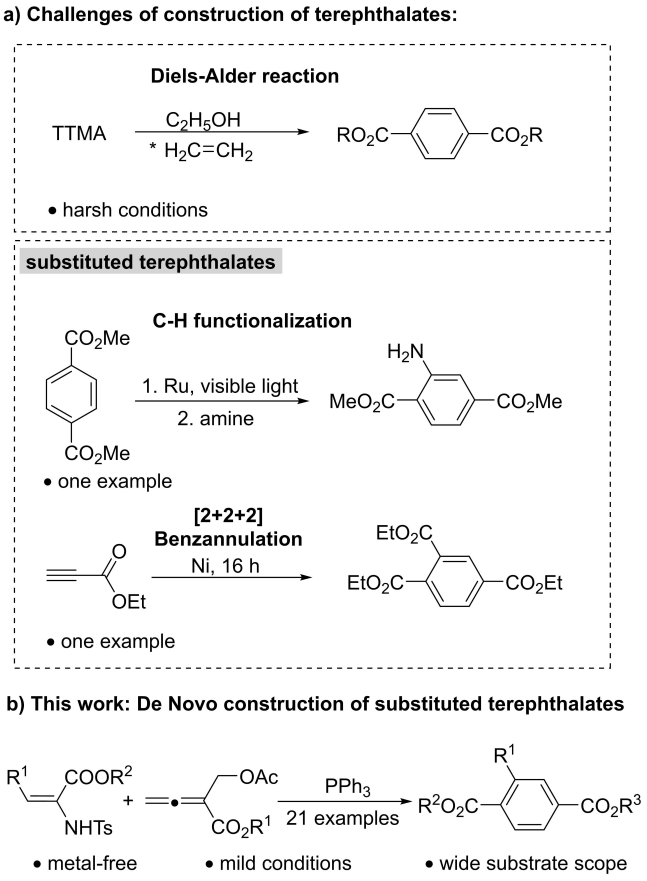


Figure 1. Sampling of pharmaceuticals containing terephthalates.

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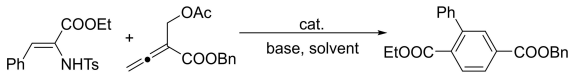
Scheme 1. Overview of the work.

pattern still limit the generality. The pronounced limitation of these strategies, coupled with the increasing emphasis on sustainability, necessitates the development of inclusive green approaches toward the construction of polysubstituted aromatic compounds.[10] From the viewpoint of diversity-oriented synthesis, developing fundamentally distinct methods for the construction of terephthalates in efficient but simple manner is still in great demand as we deem necessary, but remains a significant challenge.

Despite phosphine-catalyzed annulation of allenoates has become one of the most efficient strategy for construction of cyclic compounds,[11] only one benzannulation event occurred in 41% yield when involving β′-acetoxy allenotes.[12] As readily accessible feedstocks, enamines often serve as versatile reactive intermediates for granting a direct access to cyclization products due to their intrinsically tempered nucleophilicity.[13] Since a benzannulation-reaction-based approach to generate substituted terephthalates has not been exploited thus far, continuing our ongoing research utilizing the reactivity β′-acetoxy allenoates,[14] we designed a convenient alternative protocol for the de novo construction of terephthalate architecture via phosphine catalyzed benzannulation reaction of enamines and allenoates (Scheme 1b). Notably, terephthalate molecules possessing diverse ester groups, which are otherwise challenging to synthesize, could be precisely prepared under our reaction condition. In addition, this elegant strategy constitute a credible transform leading to biphenyl skeletons, versatile building blocks in synthetic organic chemistry.

Exploratory studies began with the [4+2] annulation of ethyl (Z)-2-((4-methylphenyl)sulfonamido)-3-phenylacrylate (**1a**) and benzyl 2-(acetoxymethyl)buta-2,3-dienoate (**2a**) as model substrates(Table 1). Expectedly, preliminary assessment with typical phosphine was found to be very encouraging, which produced the desired 5- benzyl 2-ethyl [1, 1′-biphenyl]-2, 5-dicarboxylate (**3aa**) in moderate yield when the reaction completed within five hours (entry 1). Brief bases screening only showcased unconspicuous advances (entries 1–5), and Na_2CO_3 proved to be the best choice for this domino process, giving **3aa** with 68% yield. Some solvents were attempted, whereas toluene was still the optimum solvent (entry 3 vs entries 6–10). Then meticulous optimization was continued and a pronounced improvement was found in yield under the catalysis of PPh_3 after extensive

Table 1. Screening conditions of the reaction.[a]



1a 2a 3aa

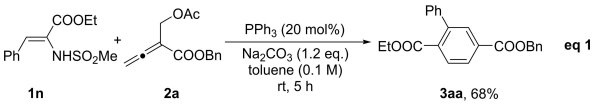
entry	cat.[mmol]	base	solvent	yield [%][b]
1	Ph_2PEt	Cs_2CO_3	Toluene	62
2	Ph_2PEt	K_2CO_3	Toluene	43
3	Ph_2PEt	Na_2CO_3	Toluene	68
4	Ph_2PEt	LiOH	Toluene	58
5	Ph_2PEt	K_3PO_4	Toluene	66
6	Ph_2PEt	Na_2CO_3	*p*-xylene	13
7	Ph_2PEt	Na_2CO_3	Mesitylene	53
8	Ph_2PEt	Na_2CO_3	Trichlorotoluene	22
9[c]	Ph_2PEt	Na_2CO_3	DMSO	47
10[c]	Ph_2PEt	Na_2CO_3	DMF	–
11	$(4\text{-}FC_6H_4)_3P$	Cs_2CO_3	Toluene	44
12	$(4\text{-}CH_3OC_6H_4)_3P$	Cs_2CO_3	Toluene	17
13	$PhEt_2P$	Cs_2CO_3	Toluene	36
14	Bu_3P	Cs_2CO_3	Toluene	28
15	PPh_3	Cs_2CO_3	Toluene	67
16	**PPh_3**	**Na_2CO_3**	**Toluene**	**75**
17	PPh_3	–	Toluene	19
18	–	Cs_2CO_3	Toluene	5

[a] Reaction conditions, unless otherwise noted:**1a** (0.1 mmol), **2a** (0.12 mmol), phosphine (0.02 mmol) and base (0.12 mmol) in 1 mL toluene were stirred at room temperature for 5 h.
[b] Isolated yield.
[c] The reaction time extended to 24 h.

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screening of organophosphines (entries 11–15). Building on the previous experimental results, we were prompted to consider Na_2CO_3 in place of Cs_2CO_3, which increased the yield to 75% successfully (entry 16). Control experiments indicated that phosphine catalyst and base are both indispensable in this transformation (entries 17 and 18).

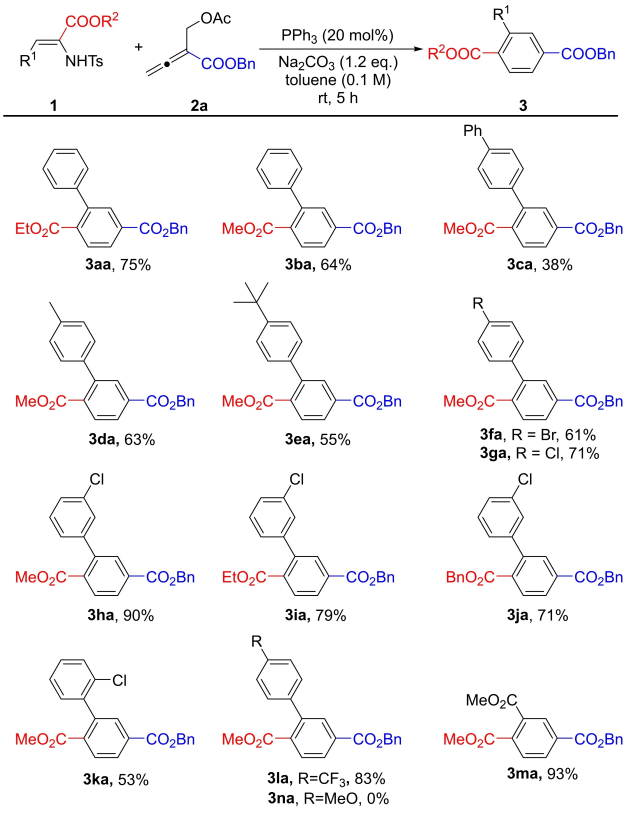
Having identified the optimal reaction conditions, we sought to investigate the diversity of this reaction (Table 2). By switching to aryl ring substituted enamine, moderate conversion was acquired (**3 ba**). Due to the steric hindrance of ester groups substituted enamines, replacing the substitution pattern with aliphatic substitution on the aryl ring turned out to be competent, while the aryl substituted derivatives showed lower reactivity toward this transformation (**3 ca**, **3 da** and **3 ea**). It is worth noting that the presence of halogen substituents on the aryl rings has a considerable impact on the efficiency of this reaction, furnished the corresponding products (**3 fa**–**3 ka**) with synthetically useful outcomes. Although the yield of the *ortho*-chloro substituted product **3 ka** is lower than the *meta* and *para* substituential products **3 ga** and **3 ha**, the *ortho*-halogen substituted aromatic ring system is relatively difficult to be prepared by other methods. Substituent including strong electron withdrawing group CF_3 on the aryl ring of substrate **1** was also well tolerated, furnishing the corresponding **3 la** in high yields (83%). Ester-substituted enamine **1 m** was also compatible for this reaction, and generated **3 ma** in excellent yield (93%). Unfortunately, when compound **1 n** with *p*-OMe substitutent on the aromatic ring was used as the substrate, the benzannulation product **3 na** couldn't be obtained. The attempts to try other enamines, e. g. alkyl substituted enamines, aryl substituted enamines with SMe, CF_3, NO_2, or CN on the aromatic ring, all failed, because the starting material couldn't be synthesized according to the known methods.[15]



When protecting group was changed into methanesulfonyl, **3 aa** could be still generated in 68% yield (eq 1). The structure was initially deduced from the NMR spectral data, and next was unambiguously confirmed by single crystal X-ray diffraction analysis (Table 2).[16]

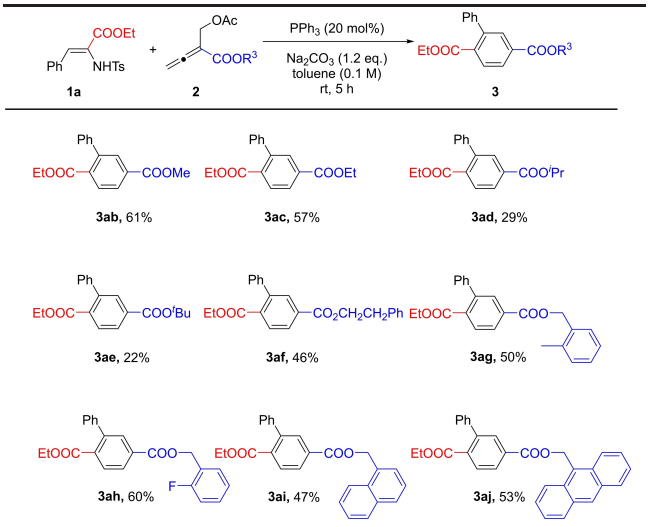
Regarding the compatibility of substituted allenoates, steric repulsion of ester groups on allenoates conspicuously influenced the chemoselectivities (Table 3). Sterically encumbered substrates, such as isopropyl and tert-butyl tend to perform poorly as the

Table 2. Substrate scope with respect to the enamines.[a,b]



[a] Reaction conditions: **1** (0.1 mmol), **2 a** (0.12 mmol), PPh_3 (0.02 mmol) and Na_2CO_3 (0.12 mmol) in 1 mL toluene were stirred at room temperature for 5 h.
[b] Isolated yields.

Table 3. Substrate scope with respect to the β′-acetoxy allenoates.[a,b]



[a] Reaction conditions: **1 a** (0.1 mmol), **2** (0.12 mmol), PPh_3 (0.02 mmol) and Na_2CO_3 (0.12 mmol) in 1 mL toluene were stirred at room temperature for 5 h.
[b] Isolated yields.

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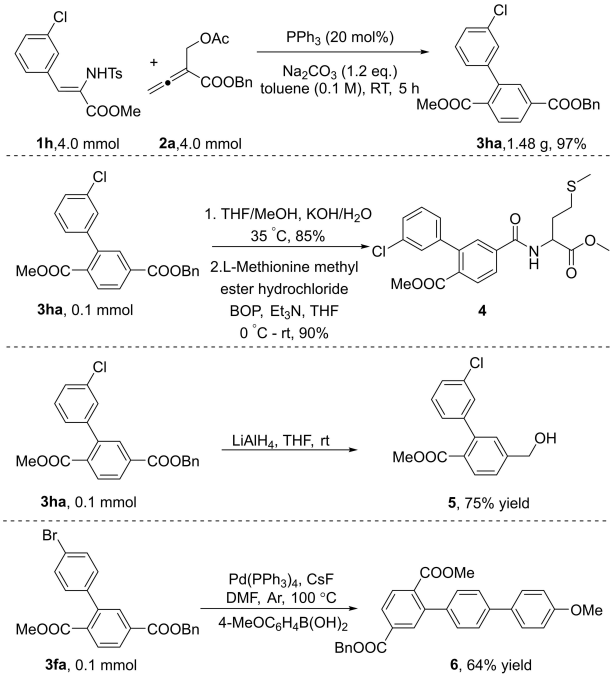
hinderance increases (**3 ab**–**3 ae**). Linear, branched, or aryl-bearing substitution on allenoates, all underwent the annulation reaction successfully to afford the corresponding products in acceptable yields (**3 ab**–**3 af**). Pleasingly, the catalytic protocol also proceeded smoothly when embedding *o*-fluorobenzyl, *o*-methylbenzyl, 1-naphthyl and 9-anthryl in substrates **2**, reflecting the mildness and high efficiency of this protocol (**3 ag**–**3 aj**).

Since the substrate generality has been well-established, we thus focused on the further illustration of the synthetic utilities. Gram experiment was performed prior to the late-stage diversification, to our delight, nearly quantitative yield was achieved under standard conditions (Scheme 2). It should be mentioned that presumably steric hindrance led to the regioselective products both in the hydrolysis and reduction of **3 ha**. Specifically, treatment of **3 ha** under inorganic basic conditions led to hydrolyzed product. The following coupling of the corresponding acid to L-methionine methyl ester hydrochloride enabled access to **4**, which can be further transformed to farnesyltransferase inhibitors.[17] Additionally, the reduction proceeded well when $LiAlH_4$ was employed, giving product **5** in 75% yield. Furthermore, 5-benzyl 2-methyl 4′-bromo-[1,1′-biphenyl]-2,5-dicarboxylates (**3 ha**) allowed for post-transformation modification and afforded the Suzuki-coupling product **6**.

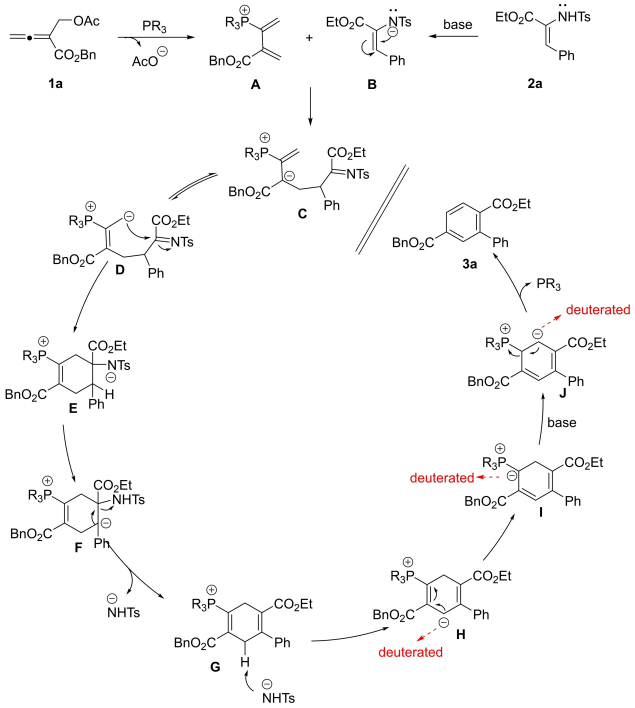
Deuterium experiment was conducted to elucidate the further mechanism of this reaction (see the Supporting Information for more details). All else being equal, 20 equiv. of D_2O was introduced into the reaction system. Incorporation of deuterium in the corresponding positions was observed, clearly indicating that intermediates with the carbanion involved in the benzannulation reaction.

Supported by preliminary mechanistic studies combined with precedent literatures, a tentative mechanism for the current benzannulation reaction is proposed (Scheme 3). 1,2-elimination of the acetate group followed by the phosphine's nucleophilic attack of phosphine at allenoate **1 a** released the intermediate **A**, which exhibited electrophilicity and next served as a four-carbon reacting partner for the catalytic annulation reaction. With the help of base, enamine interacted with **A** leading to intermediate **C**. Then, intermediate **G** was delivered undergoing 1,2-elimination followed by intramolecular addition and proton transfer process. Finally, the aromaticity is constructed after deprotonation and isomerization to afford the benzannulation product **3 a**, along with the regeneration of phosphine catalyst.

In summary, we reported on the discovery of a strategically distinct approach to substituted terephthalates via domino benzannulation of β′-acetoxy allenoates and enamines by environmentally friendly conditions and commercially available phosphine catalyst, which can be drove with synthetically useful yields. This protocol features a broad substrate scope and attractively mild conditions. Additionally, the late-stage functionalization illustrated the synthetic utility.



Scheme 2. Gram-scale synthesis and late-stage diversification.



Scheme 3. Plausible mechanism of the reaction of **1 a** and **2 a**.

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Experimental Section

General Procedure for the Synthesis of 3: To a 10 mL glass vial was added enamins **1** (0.12 mmol, 1.2 equiv.), allenoates **2** (0.1 mmol, 1.0 equiv.), Na_2CO_3 (0.12 mmol, 1.2 equiv.), PPh_3 (0.02 mmol, 0.2 equiv.) and toluene (1 mL). The resulting suspension was stirred at room temperature for 5 hours. After the removal of the solvent under reduced pressure, the residue was purified by flash column chromatography (petroleum ether: ethyl acetate = 5:1) to afford products **3**.

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